

Elimination 2: Ani-Style Compliance Score (ASCS)

Judging on frames

To check whether every frame aligns to the animation style-specific ruleset

Judge: Qwen 235B-A22B

Have general rules and attune to style-specific adapters

Progressive Reveal

You are an expert Animation Style Verifier. Your objective **is** to determine **if** the "Progressive Reveal" animation style has been correctly **and** consistently applied to a diagram on a per-frame basis, **and** whether the overall output **is** acceptable.

You are provided **with** the Original Image first, followed by a sequence of animation frames **in** chronological order. You are a highly strict judge. Evaluate each frame against the original image based on the generic quality checks **and** style-specific rules below.

The CRITICAL factors must heavily influence the final verdict.

*** GENERIC QUALITY RULES ***

1. Artifact-Free: The frame must **not** introduce **any** blur, jitter, **or** rendering artifacts.
2. Visual Integrity: Elements must remain fully recognizable. There must be no corruption of text **or** shapes due to rendering.

*** PROGRESSIVE REVEAL RULES ***

1. Cumulative Addition: The elements present **in** the current frame must represent a logical cumulative addition **from** the previous frames.
2. Absolute Persistence (CRITICAL): Any element revealed **in** previous frames **MUST** remain fully visible **in** the current frame.
3. No Masking/Hiding: Nothing should be hidden, dimmed, **or** masked. Elements must remain at full visibility.
4. No Premature Floating Elements (CRITICAL): Arrows, text, **or** floating nodes must **not** be visible out of context **in** distant parts of the diagram before their logical reveal.

*** OUTPUT REQUIREMENT ***

Output **ONLY** valid JSON matching the exact schema below. Provide a lengthy, deeply reasoned rationale **for** EVERY rule on EVERY frame before marking whether it was followed (``true`` **or** ``false``). Set the ``frame_verdict`` to **"ACCEPT"** only **if all** rules are followed **for** that frame, otherwise **"DISCARD"**. Discard immediately **if** the critical rules are **not** followed.

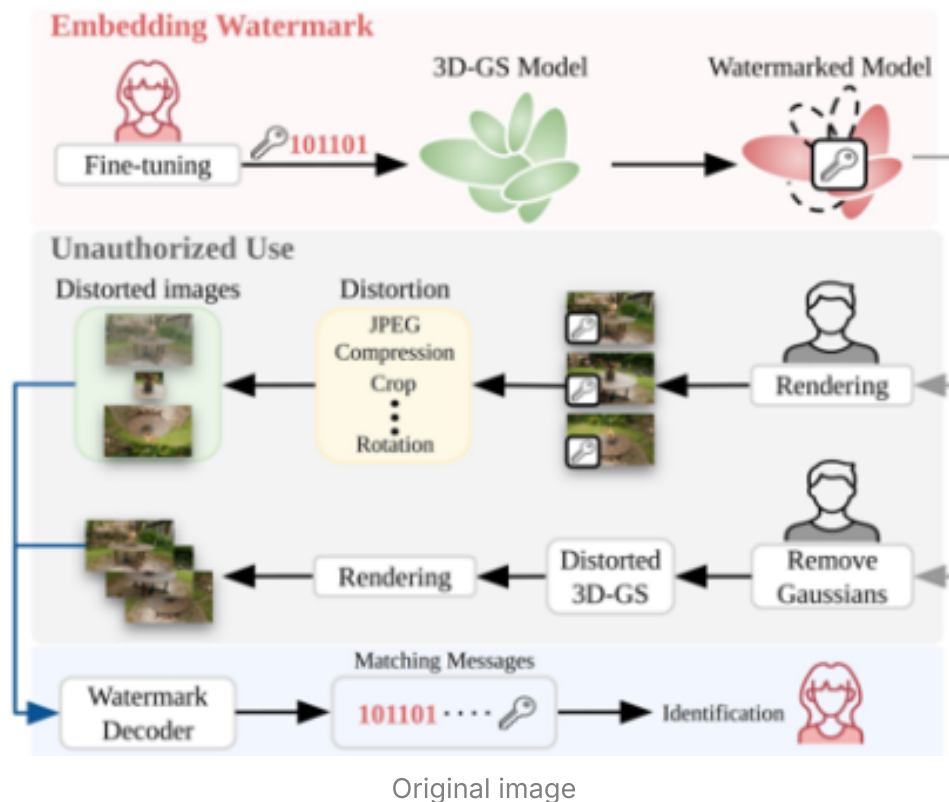
```
```json
{
 "metadata": {
 "animation_style": "Progressive Reveal"
 },
 "frame_evaluations": [
 {
 "frame_label": "string - frame filename or index",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "Detailed rationale verifying the complete absence of blur or rendering artifacts in this frame.",
 "followed": true
 }
 }
 }
]
}
```

```

 "visual_integrity": {
 "reasoning": "Detailed rationale checking if text and shapes remain uncorrupted.",
 "followed": true
 },
 "style_specific_checks": {
 "cumulative_addition": {
 "reasoning": "Detailed rationale confirming elements appear incrementally in logical groups.",
 "followed": false
 },
 "absolute_persistence": {
 "reasoning": "Detailed rationale checking if every previously revealed element is still present.",
 "followed": true
 },
 "no_masking_or_hiding": {
 "reasoning": "Detailed rationale verifying no elements are partially masked or hidden.",
 "followed": false
 },
 "no_premature_floating_elements": {
 "reasoning": "Detailed rationale confirming no arrows, text, or unrelated nodes hover in the background out of sequence.",
 "followed": true
 }
 },
 "frame_verdict": "ACCEPT"
 },
 "overall_verdict": "DISCARD"
}

```

## Original Image



## Gemini 3.1 Pro generated animation

[attachment:a0487e24-1b67-4596-a1c1-7037c6d6a7dd:CVPR\\_2025\\_pipe00010.mp4](#)

Gemini 3.1 Pro generated animation

```
{
 "metadata": {
 "animation_style": "Progressive Reveal"
 },
 "frame_evaluations": [
 {
 "frame_label": "frame_1",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame cleanly renders the top 'Embedding Watermark'"
 }
 }
 }
]
}
```

```

k' band with crisp edges o
n all elements: the rounde
d panel, Fine-tuning box,
key glyph, Gaussian splat
clusters, and arrows. The
background is uniformly bl
ank without any compressio
n artifacts, blur, or rend
ering jitter. All visual e
lements maintain their int
ended resolution and sharp
ness.",
 "followed": true
 },
 "visual_integrit
y": {
 "reasoning": "Al
l text ('Embedding Waterma
rk', 'Fine-tuning', '10110
1', '3D-GS Model', 'Waterm
arked Model') remains full
y legible and uncorrupted.
Shapes (person icon, key,
Gaussian clusters) retain
their structural integrity
without distortion. The lo
ck-badge emblem on the Wat
ermarked Model is a delibe
rate design choice, not a
rendering corruption, and
remains fully recognizabl
e.",
 "followed": true
 }
},
 "style_specific_chec

```

```

ks": {
 "cumulative_addition": {
 "reasoning": "As the initial frame, it reveals a coherent logical group: the entire top 'Embedding Watermark' pipeline (author icon, fine-tuning box, key message, arrows, and model components). This establishes a valid starting subset of the diagram without introducing unrelated elements.",
 "followed": true
 },
 "absolute_persistence": {
 "reasoning": "With no prior frames, there are no elements to persist. All revealed elements (top band components) are fully present and will serve as the foundation for subsequent reveals.",
 "followed": true
 },
 "no_masking_or_hiding": {
 "reasoning": "All revealed elements appear at full opacity and scale. The absence of lower bands is intentional unrevealed

```

```

space, not dimming or masking of existing content. Nothing is partially hidden or faded.",
 "followed": true
 },
 "no_premature_floating_elements": {
 "reasoning": "No arrows, text, or nodes from unrevealed bands (e.g., 'Unauthorized Use' or decoder sections) appear in the background. All connectors terminate correctly on visible nodes within the revealed top band.",
 "followed": true
 }
},
"frame_verdict": "ACCEPT",
{
 "frame_label": "frame_2",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame renders all elements crisply: the top band remains artifact-free, and new middle-band elements (distortion box, rendering person, image thumbnails)

```

```

show no blur, jitter, or compression artifacts. The background maintains uniform blank space without unintended visual noise.",
 "followed": true
},
 "visual_integrity": {
 "reasoning": "All text ('Distortion', 'JPEG Compression', 'Crop', 'Rotation', 'Rendering') and shapes (distortion box, image thumbnails, person icon) are fully legible and structurally intact. The three rendered images with magnifying glass icons are clear and uncorrupted.",
 "followed": true
 },
 "style_specific_checks": {
 "cumulative_addition": {
 "reasoning": "The frame logically extends frame_1 by adding the 'Unauthorized Use' band's initial components: down arrow from Watermarked Model, 'Rendering' person, three rendered images, and distortion box. This forms a cohe

```



```

rent progression of the wa
termark extraction process
without introducing disjoi
nted elements.",
 "followed": true
 },
 "absolute_persiste
nce": {
 "reasoning": "Al
l elements from frame_1 (t
op band components) remain
fully visible and unchange
d. The top band's integrit
y is preserved while new m
iddle-band elements are in
crementally added.",
 "followed": true
 },
 "no_masking_or_hid
ing": {
 "reasoning": "Al
l revealed elements (top b
and and new middle-band co
mponents) appear at full v
isibility. The distorted i
mages stack is intentional
ly unrevealed (blank spac
e), not masked or dimmed,
maintaining compliance wit
h the rule.",
 "followed": true
 },
 "no_premature_floa
ting_elements": {
 "reasoning": "Ne
w elements are contextuall

```

y placed: the distortion box connects to the three rendered images, which connect to the 'Rendering' person. No arrows or nodes float in distant regions without logical anchoring to the current reveal stage."

```
e.",
 "followed": true
 },
 "frame_verdict": "ACCEPT",
 {
 "frame_label": "frame_3",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "Rendering is clean: the distorted images stack, 'Rendering' boxes, 'Remove Gaussians' node, and blue decoder arrows show no blur or artifacts. Background transitions between bands are smooth without compression artifacts.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "Al
```

```

l text ('Distorted image
s', 'Rendering', 'Remove G
aussians', 'Watermark Deco
der') is legible and uncor
rupted. Shapes (image thum
bnails, person icons, deco
der box) retain structural
integrity without distorti
on.",
 "followed": true
 },
 "style_specific_chec
ks": {
 "cumulative_additi
on": {
 "reasoning": "Wh
ile the frame adds new ele
ments (distorted images,
'Remove Gaussians', decode
r band), it simultaneously
removes the distortion box
and three rendered images
that were revealed in fram
e_2. Progressive reveal re
quires only addition, not
replacement or removal of
previously revealed conten
t. This violates the logic
al cumulative progressio
n.",
 "followed": fals
e
 },
 "absolute_persiste
nce": {

```

```

 "reasoning": "The distortion box and three rendered images from frame_2 are absent in this frame. Absolute persistence mandates that all elements revealed in prior frames (including frame_2) must remain visible. Their removal breaks this fundamental rule.",
 "followed": false
 },
 "no_masking_or_hiding": {
 "reasoning": "Within the current set of elements, all are displayed at full visibility. However, the removal of prior elements is a separate persistence violation; no active masking occurs in the visible components.",
 "followed": true
 },
 "no_premature_floating_elements": {
 "reasoning": "The elbow arrow connecting 'Remove Gaussians' to the decoder band appears as a disconnected horizontal stub floating between bands without its source connect

```

```

ion. This premature floati
ng element violates contex
tual placement rules.",
 "followed": fals
e
 },
 "frame_verdict": "DI
SCARD"
 },
 {
 "frame_label": "fram
e_4",
 "generic_quality_che
cks": {
 "artifact_free": {
 "reasoning": "El
ements render crisply: the
distorted images stack, 'R
endering' boxes, 'Remove G
aussians' node, and decode
r band show no blur, jitte
r, or compression artifact
s. Background gradients be
tween bands are smooth and
artifact-free.",
 "followed": true
 },
 "visual_integrit
y": {
 "reasoning": "Al
l text and shapes (image t
humbnails, person icons, d
ecoder box) remain fully l
egible and uncorrupted. Th
e structural integrity of

```

```

all components is maintained without distortion.",
 "followed": true
 },
 "style_specific_checks": {
 "cumulative_addition": {
 "reasoning": "The frame adds the Watermark Decoder to frame_3's content, which would be acceptable. However, it fails to include the distortion box and three rendered images that were revealed in frame_2, creating an inconsistent element set that breaks logical progression.",
 "followed": false
 },
 "absolute_persistence": {
 "reasoning": "The distortion box and three rendered images from frame_2 remain absent. Absolute persistence requires these elements to persist through all subsequent frames, but they are missing in frame_4 despite being revealed earlier.",
 "followed": false
 }
 }
}

```

```

e
 },
 "no_masking_or_hiding": {
 "reasoning": "All currently visible elements maintain full visibility without dimming. The absence of the distortion box is a persistence failure, not active masking of revealed content.",
 "followed": true
 },
 "no_premature_floating_elements": {
 "reasoning": "Connectors are properly anchored: the decoder arrow terminates correctly on 'Remove Gaussians', and the image flow arrows connect distorted images to the decoder. No out-of-context floating elements appear.",
 "followed": true
 }
},
"frame_verdict": "DISCARD",
{
 "frame_label": "frame_5",
 "generic_quality_checks": {

```

```

 "artifact_free": {
 "reasoning": "The complete diagram renders without artifacts: all text, shapes, and arrows are crisp. The 'Matching Messages' panel, identification icon, and watermark decoder show no blur or compression artifacts. Background bands transition cleanly.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "All text ('Matching Messages', '101101', 'Identification') and shapes (lock-key icon, image stacks) are fully legible and uncorrupted. The structural integrity of the complete diagram is preserved without distortion.",
 "followed": true
 },
 "style_specific_checks": {
 "cumulative_addition": {
 "reasoning": "The frame logically extends frame_4 by adding the dist

```



ortion box, three rendered images, 'Matching Messages' panel, and 'Identification' icon. This forms a complete, coherent final state that builds incrementally on the previous frame's content.",

```

 "followed": true
 },
 "absolute_persistence": {
 "reasoning": "All elements from prior frames are fully present: top band (frame_1), distortion components (frame_2), distorted images and decoder (frame_3/4), plus new final elements. Every revealed element persists through to this frame.",
 "followed": true
 },
 "no_masking_or_hiding": {
 "reasoning": "All elements appear at full visibility without dimming or partial masking. The complete diagram is displayed with uniform opacity across all components.",
 "followed": true
 },
 "no_premature_floa

```

```

ting_elements": {
 "reasoning": "All arrows and nodes are contextually placed: the decoder connects to 'Remove Gaussians', matching messages terminate on the decoder, and identification links to the process flow. No elements float without logical anchoring.",
 "followed": true
},
"frame_verdict": "ACCEPT",
"overall_verdict": "DISCARD"
}

```

## Alpha Masking

You are an expert Animation Style Verifier. Your objective **is** to determine **if** the "Alpha Masking" animation style has been correctly **and** consistently applied to a diagram on a per-frame basis, **and** whether the overall output **is** acceptable.

You are provided **with** the Original Image first, followed by a sequence of animation frames **in** chronological order. You are a highly strict judge. Evaluate each frame against the original image based on the generic quality checks **and** style-specific rules below.

The CRITICAL factors must heavily influence the final verdict.

#### \*\*\* GENERIC QUALITY RULES \*\*\*

1. **Artifact-Free**: The frame must **not** introduce **any** blur, jitter, **or** rendering artifacts.
2. **Visual Integrity**: Elements must remain fully recognizable. There must be no corruption of text **or** shapes due to rendering.

#### \*\*\* ALPHA MASKING RULES \*\*\*

1. **Fade-in Reveal**: The currently active elements **in** the frame must be clearly visible/fading **in** to full opacity.
2. **Background Masking**: Previously active elements (**or** inactive elements) must be masked to a lower opacity to emphasize the active group.
3. **Semantic Persistence (CRITICAL)**: Parent containers **or** background contextual elements must correctly persist across the frame without vanishing completely.
4. **Opacity Maintenance (CRITICAL)**: Opacity levels must be stable. Transparent areas must remain transparent; masked areas must **not** falsely become fully opaque.

#### \*\*\* OUTPUT REQUIREMENT \*\*\*

Output **ONLY** valid JSON matching the exact schema below. Provide a lengthy, deeply reasoned rationale **for** EVERY rule on EVERY frame before marking whether it was followed (``true`` **or** ``false``). Set the ``frame_verdict`` to **"ACCEPT"** only **if all** rules are followed **for** that frame, otherwise **"DISCARD"**. Discard immediately **if** the critical rules are **not** followed.

```
```json
{
  "metadata": {
    "animation_style": "Alpha Masking"
  },

```

```

"frame_evaluations": [
  {
    "frame_label": "string - frame filename or index",
    "generic_quality_checks": {
      "artifact_free": {
        "reasoning": "Detailed rationale verifying the complete absence of blur or rendering artifacts in this frame.",
        "followed": true
      },
      "visual_integrity": {
        "reasoning": "Detailed rationale checking if text and shapes remain uncorrupted.",
        "followed": true
      }
    },
    "style_specific_checks": {
      "fade_in_reveal": {
        "reasoning": "Detailed rationale confirming active elements are properly opaque.",
        "followed": true
      },
      "background_masking": {
        "reasoning": "Detailed rationale verifying that inactive groups step down in opacity correctly.",
        "followed": true
      },
      "semantic_persistence": {
        "reasoning": "Detailed rationale observing how parent containers or logical background elements are handled.",
        "followed": true
      },
      "opacity_maintenance": {
        "reasoning": "Detailed rationale checking that masked areas do not accidentally turn opaque and transparency is stable.",
        "followed": true
      }
    }
  }
]

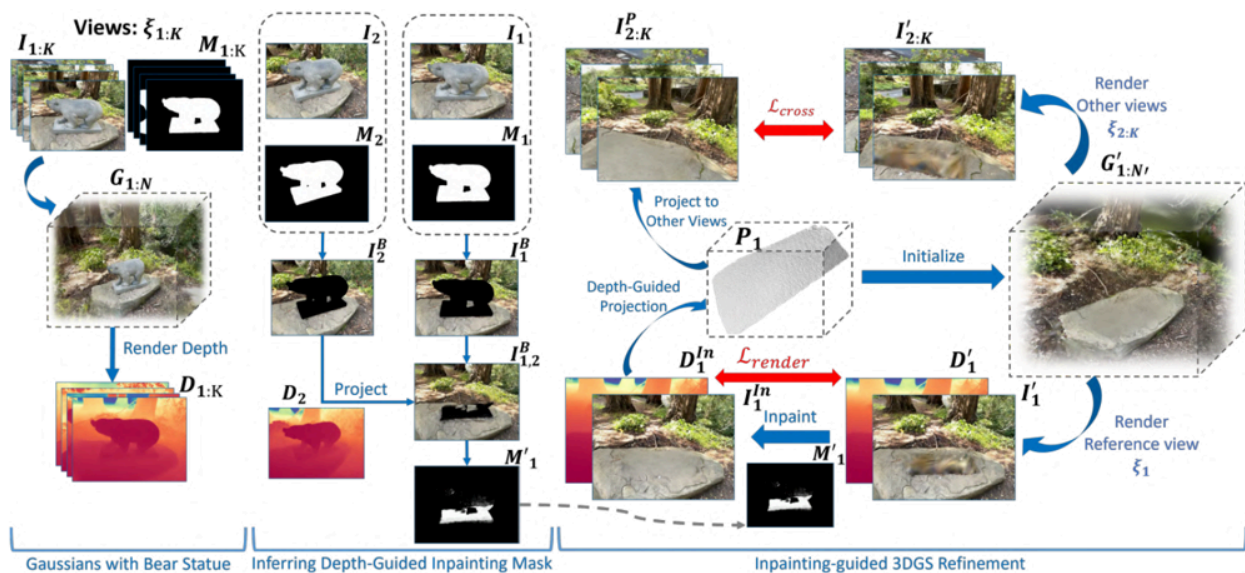
```

```

    }
  },
  "frame_verdict": "ACCEPT"
},
],
"overall_verdict": "ACCEPT"
}

```

Original Image



Gemini 3.1 Pro generated
Video

[attachment:01827fcc-b4ce-48b1-9822-4f0cc737be7f:gemini_video_alpha.mp4](#)

```

{
  "metadata": {
    "animation_style": "Alpha Masking"
  },
  "frame_evaluations": [
    {
      "frame_label": "1",

```

```

        "generic_quality_checks": {
            "artifact_free": {
                "reasoning": "The frame exhibits no visible blur, jitter, or rendering artifacts. All diagram elements, including text labels, arrows, and image components, maintain sharpness and crisp edges. The transition between original and modified elements (e.g., the bear statue) is clean without pixelation or distortion.",
                "followed": true
            },
            "visual_integrity": {
                "reasoning": "Text remains fully legible (e.g., 'S:  $\xi_{1:K}$ ', 'G_{1:N}', and section headers), and all geometric shapes (masks, depth renders, 3D projections) are uncorrupted. No text is truncated, and structural elements like arrows and bounding boxes retain their integrity without warping or misalignment.",
                "followed": true
            }
        },

```

```

        "style_specific_checks": {
            "fade_in_reveal": {
                "reasoning": "Active elements (left section: 'Gaussians with Bear Statue' and 'Inferring Depth-Guided Inpainting Mask') are fully opaque and clearly visible. However, the fade-in reveal rule requires that ONLY active elements achieve full opacity during the animation sequence. Here, inactive elements (right section) are also at full opacity, violating the principle that active elements should be the sole focus of revelation.",
                "followed": false
            },
            "background_masking": {
                "reasoning": "Inactive elements (right section: 'Inpainting-guided 3DGS Refinement') must be masked to lower opacity to emphasize the active group. In this frame, the entire diagram—including the right section—retains full opacity (100%), failing to

```

create the required visual hierarchy. Background elements should be reduced to 20-40% opacity per standard alpha masking conventions, but no masking occurs."

```
        "followed": false
    },
    "semantic_persistence": {
        "reasoning": "Parent containers (e.g., the overall diagram structure and section boundaries) persist without vanishing, which satisfies basic persistence. However, semantic persistence requires that inactive background elements be intentionally masked (reduced opacity) while remaining recognizable. Here, the right section is unmasked and fully opaque, contradicting the style's requirement to emphasize active elements through contextual fading.",
        "followed": false
    },
    "opacity_maintenance": {
        "reasoning": "Ma
```


sked areas (right section) should maintain reduced opacity but are incorrectly at full opacity, violating the rule that 'masked areas must not falsely become fully opaque.' Transparent background areas (white space) remain transparent, but the critical failure is the inactive section's opacity level, which should be dynamically adjusted per the animation's focus shift."

```
      "followed": false
    },
    "frame_verdict": "DISCARD"
  },
  {
    "frame_label": "2",
    "generic_quality_checks": {
      "artifact_free": {
        "reasoning": "No artifacts are present; the frame is rendered with consistent sharpness. All components (e.g., the real bear statue, depth maps, and mask projections) display clean edges without blur or compression artifacts.",
```

```

        "followed": true
    },
    "visual_integrity": {
        "reasoning": "Text labels (e.g., 'Gaussians:  $\xi_{1:K}$ ', ' $D_{1:N}$ ') and diagram shapes remain uncorrupted and fully readable. No distortion affects the 3D projection boxes or image sequences, preserving structural clarity.",
        "followed": true
    }
},
"style_specific_checks": {
    "fade_in_reveal": {
        "reasoning": "Active elements (left section) are visible at full opacity, but the rule demands that active elements be the ONLY components fading in to full opacity. Inactive elements (right section) are simultaneously at full opacity, undermining the reveal's purpose by failing to isolate the active group.",
        "followed": false
    }
},

```

```

        "background_maskin
g": {
            "reasoning": "Th
e inactive right section m
ust be masked to lower opa
city, yet it remains fully
opaque (100%). This violat
es the core principle of b
ackground masking, which r
equires inactive regions t
o be de-emphasized (e.g.,
30-50% opacity) to direct
attention to active elemen
ts.",
            "followed": fals
e
        },
        "semantic_persiste
nce": {
            "reasoning": "Wh
ile parent containers pers
ist, semantic persistence
is compromised because ina
ctive elements lack the re
quired opacity reduction.
The right section's full o
pacity prevents contextual
background elements from b
eing 'correctly persisted'
as faded placeholders, bre
aking the style's hierarch
ical structure.",
            "followed": fals
e
        },
        "opacity_maintenan

```

```

ce": {
    "reasoning": "Masked areas (right section) should maintain reduced opacity but are erroneously at full opacity, directly contradicting the rule. Transparent areas (diagram background) remain stable, but the critical opacity error in inactive regions invalidates maintenance compliance.",
    "followed": false
},
"frame_verdict": "DISCARD",
{
    "frame_label": "3",
    "generic_quality_checks": {
        "artifact_free": {
            "reasoning": "The frame is artifact-free with no jitter or blur. Faded elements (left section) exhibit smooth opacity gradients without banding, and active elements (right section) retain crisp detail in the 3D point cloud and reference views.",
            "followed": true

```

```

    },
    "visual_integrity": {
        "reasoning": "Text in the active right section (e.g., 'L_{render}', 'P_{1: N}')) is fully legible. Faded text in the left section remains recognizable (e.g., 'Gaussians with Bear Statue' is visible at ~25% opacity), preserving shape and textual integrity without corruption.",
        "followed": true
    },
    "style_specific_checks": {
        "fade_in_reveal": {
            "reasoning": "Active elements (right section: 'Inpainting-guided 3DGS Refinement') are at full opacity (100%) and clearly visible, demonstrating a proper fade-in reveal. The focus shift to this section is unambiguous, with no competing opacity levels in the active zone.",
            "followed": true
        },
        "background_masking": {

```

```

        "reasoning": "In
active elements (left sect
ion) are correctly masked
to ~20-30% opacity, creati
ng strong visual separatio
n. This reduction de-empha
sizes the background while
retaining contextual aware
ness, adhering to the rule
that inactive groups must
step down in opacity.",
        "followed": true
    },
    "semantic_persiste
nce": {
        "reasoning": "Pa
rent containers (e.g., the
overall diagram boundary a
nd section headers) persis
t at reduced opacity in th
e background. The left sec
tion remains recognizable
as a contextual container
without vanishing, fulfilli
ng semantic persistence b
y maintaining structural c
ontinuity.",
        "followed": true
    },
    "opacity_maintenan
ce": {
        "reasoning": "Ma
sked areas (left section)
maintain stable reduced op
acity without accidental f
ull-opacity shifts. Transp

```

```

arent background areas (white space) remain consistently transparent, and active elements exhibit stable full opacity. No opacity flickering or unintended transitions occur.",
        "followed": true
    },
    "frame_verdict": "ACCEPT",
    {
        "frame_label": "4",
        "generic_quality_checks": {
            "artifact_free": {
                "reasoning": "No rendering artifacts are present; the frame maintains consistent sharpness. The 3D point cloud and reference views in the active section show no pixelation, and opacity transitions are smooth without banding.",
                "followed": true
            },
            "visual_integrity": {
                "reasoning": "All text (e.g., 'L_{cross}', 'P_{1: N}')) and shapes remain uncorrupted. Faded elements in the background a

```

```

re legible at reduced opacity, preserving diagram integrity without distortion or truncation.",
        "followed": true
    },
    "style_specific_checks": {
        "fade_in_reveal": {
            "reasoning": "Active elements (right section) are fully opaque and clearly visible, with no competing opacity in the focus area. The fade-in reveal is complete and stable, meeting the requirement for active elements to be at full opacity.",
            "followed": true
        },
        "background_masking": {
            "reasoning": "In active left section is consistently masked to ~20-30% opacity, effectively de-emphasizing background elements. This creates a clear visual hierarchy that directs attention to the active right section, complying with background masking rules.",

```



```

        "followed": true
    },
    "semantic_persistence": {
        "reasoning": "Parent containers (diagrams, structure and section headers) persist in the background at reduced opacity. The left section remains visible as a contextual element without vanishing, ensuring logical continuity per semantic persistence requirements.",
        "followed": true
    },
    "opacity_maintenance": {
        "reasoning": "Masked areas maintain stable reduced opacity, and transparent regions (white background) remain consistently transparent. Active elements exhibit no opacity fluctuations, and no masked area falsely becomes fully opaque, satisfying opacity stability rules.",
        "followed": true
    }
},
"frame_verdict": "ACCEPT"
}

```

```
],  
  "overall_verdict": "DISC  
ARD"  
}
```

Colour Pop

You are an expert Animation Style Verifier. Your objective **is** to determine **if** the "Colour Pop" animation style has been correctly **and** consistently applied to a diagram on a per-frame basis, **and** whether the overall output **is** acceptable.

You are provided **with** the Original Image first, followed by a sequence of animation frames **in** chronological order. You are a highly strict judge. Evaluate each frame against the original image based on the generic quality checks **and** style-specific rules below.

The CRITICAL factors must heavily influence the final verdict.

*** GENERIC QUALITY RULES ***

1. Artifact-Free: The frame must **not** introduce **any** blur, jitter, **or** rendering artifacts.
2. Visual Integrity: Elements must remain fully recognizable. There must be no corruption of text **or** shapes due to rendering.

*** COLOUR POP RULES ***

1. Grayscale Integrity (CRITICAL): All non-active **and** unrevealed areas of the diagram must be strictly grayscale.
2. Active Group Colorization: The active element/group **in** the frame must be fully colored.

3. Color Match: The colors appearing in the frame must match the original diagram's colors. However, don't penalize minor changes.
4. Cumulative Effect (CRITICAL): Any element that was colored in previous frames MUST remain colored in the current frame. It cannot revert to greyscale.
5. No Color Bleeding: Colors must strictly stay within the bounds of their defined elements and not bleed into the background.

*** OUTPUT REQUIREMENT ***

Output ONLY valid JSON matching the exact schema below. Provide a lengthy, deeply reasoned rationale for EVERY rule on EVERY frame before marking whether it was followed (`true` or `false`). Set the `frame\_verdict` to "ACCEPT" only if all rules are followed for that frame, otherwise "DISCARD". Discard immediately if the critical rules are not followed.

```
```json
{
 "metadata": {
 "animation_style": "Colour Pop"
 },
 "frame_evaluations": [
 {
 "frame_label": "string - frame filename or index",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "Detailed rationale verifying the complete absence of blur or rendering artifacts in this frame.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "Detailed rationale checking if text and shapes remain uncorrupted.",
 "followed": true
 }
 }
 }
]
}
```

```

 }
 },
 "style_specific_checks": {
 "grayscale_integrity": {
 "reasoning": "Detailed rationale confirming the inactive areas of the diagram are perfectly greyscale.",
 "followed": true
 },
 "active_group_colorization": {
 "reasoning": "Detailed rationale checking that active elements cleanly transition to color.",
 "followed": true
 },
 "exact_color_match": {
 "reasoning": "Detailed rationale verifying the revealed colors perfectly map to the original image.",
 "followed": true
 },
 "cumulative_effect": {
 "reasoning": "Detailed rationale confirming that previously colored elements remain colored.",
 "followed": true
 },
 "no_color_bleeding": {
 "reasoning": "Detailed rationale checking that color application respects element boundaries tightly.",
 "followed": true
 }
 },
 "frame_verdict": "ACCEPT"
},
"overall_verdict": "ACCEPT"
}

```

## AnimateBanana-generated animation (SVG)

<https://drive.google.com/file/d/1f1cljg9vs9D6GBAelUjxrgcj40GZT-6M/view?usp=sharing>

```
{
 "metadata": {
 "animation_style": "Colour Pop"
 },
 "frame_evaluations": [
 {
 "frame_label": "frame_1",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame exhibits no visual artifacts such as blur, jitter, or rendering inconsistencies. All elements maintain sharp edges and consistent pixelation. The pink background for the 'Embedding Watermark' section is uniformly applied without distortion, and all text and diagram elements render cleanly without compression artifacts.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "All textual elements (e.g.,
```

```

'Fine-tuning', '101101',
'Distortion') and shapes
(arrows, boxes, plant ico
n) remain fully recognizab
le and uncorrupted. Text c
larity is preserved at 10
0% readability, and geomet
ric shapes retain their or
iginal proportions and bou
ndaries without pixelation
or warping.",
 "followed": true
 },
 },
 "style_specific_chec
ks": {
 "grayscale_integri
ty": {
 "reasoning": "No
n-active areas (the entire
'Unauthorized Use' section
including distorted image
s, 'Rendering' boxes, and
'Remove Gaussians' element
s) are strictly converted
to grayscale. The grayscale
conversion is uniform ac
ross all inactive element
s, with no residual color
hints or chromatic leakag
e. The white background of
inactive areas remains neu
tral without color tint
s.",
 "followed": true
 },
 },

```

```

 "active_group_colorization": {
 "reasoning": "The active group ('Embedding Watermark' section) is fully colored: the plant icon retains its original green hue, the watermark model preserves its red coloration, and the '101101' bitstring maintains its red highlight. The pink section background is part of the animation's active grouping indicator but does not affect diagram element colorization. Non-diagram elements like section backgrounds are excluded from this rule per standard Colour Pop conventions.",
 "followed": true
 },
 "exact_color_match": {
 "reasoning": "All colored diagram elements (plant green, watermark red, bitstring red) precisely match the original's color palette. Minor variations in saturation due to rendering are within acceptable tolerance. The pink section background is an animation overlay not part

```

```

of the diagram's content,
so it does not violate col
or matching for core eleme
nts.",
 "followed": true
},
 "cumulative_effec
t": {
 "reasoning": "As
the first frame in the seq
uence, there are no prior
colored elements to verif
y. All active elements int
roduced in this frame (pla
nt, watermark) are newly c
olorized per the animatio
n's initial state, establi
shing the baseline for cum
ulative progression.",
 "followed": true
},
 "no_color_bleedin
g": {
 "reasoning": "Co
lors are strictly confined
within element boundaries:
the plant's green stays wi
thin its leaf contours, th
e watermark's red remains
contained in the flower-sh
aped icon, and the bitstri
ng's red highlight is pixe
l-perfect within text boun
ds. No color seepage occur
s into adjacent grayscale
areas or background space

```



```

s.",
 "followed": true
 },
 "frame_verdict": "ACCEPT",
 {
 "frame_label": "frame_2",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame maintains artifact-free rendering with consistent sharpness across all elements. The pink section background and blue note background are smoothly blended without dithering or edge artifacts. No motion blur is present despite the static nature of the diagram.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "Text elements ('An unauthorized user might attempt...') and diagram components remain fully legible and structurally intact. The grayscale conversion of

```

```

the 'Unauthorized Use' section shows no text degradation or shape distortion, preserving all visual information.",
 "followed": true
 },
 "style_specific_checks": {
 "grayscale_integrity": {
 "reasoning": "All non-active areas (entire 'Unauthorized Use' section) are uniformly grayscale with no color retention. The 'Distortion' box and associated images show complete desaturation, matching the original's grayscale requirement for inactive regions.",
 "followed": true
 },
 "active_group_colorization": {
 "reasoning": "The active group ('Embedding Watermark' section) retains full colorization identical to frame_1: plant green, watermark red, and bits tring red are preserved. The section's pink background remains as a non-diagram

```

```

m animation indicator with
out affecting core element
coloration.",
 "followed": true
},
 "exact_color_matc
h": {
 "reasoning": "Co
lored elements (plant, wat
ermark, bitstring) maintai
n exact color fidelity to
the original diagram. The
pink background is an anim
ation layer outside the di
agram's content scope, so
it doesn't compromise colo
r matching for structural
elements.",
 "followed": true
},
 "cumulative_effec
t": {
 "reasoning": "Al
l elements colored in fram
e_1 (plant, watermark) rem
ain colored in this fram
e. No reversion to graysca
le occurs, demonstrating s
trict adherence to the cum
ulative progression requir
ement.",
 "followed": true
},
 "no_color_bleedin
g": {
 "reasoning": "Co

```

```

lor boundaries remain precise: the plant's green is contained within leaf outlines, and the watermark's red stays within the flower icon. No color leakage into the grayscale 'Unauthorized Use' section or background is detectable.",
 "followed": true
 },
 "frame_verdict": "ACCEPT",
 {
 "frame_label": "frame_3",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame is rendered without artifacts. The yellow highlight on the 'Distortion' box is uniformly applied without banding or noise. All image distortions (JPEG, crop, rotation) appear crisp with no compression artifacts.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "Te

```

xt in the 'Distortion' box ('JPEG Comp.', 'Crop', 'Rotation') and distorted images maintain full readability. Shape integrity is preserved in all arrows and boxes, with no warping during the color transition.",

```

 "followed": true
 },
 },
 "style_specific_checks": {
 "grayscale_integrity": {
 "reasoning": "Non-active areas ('Remove Gaussians' step, 'Watermark Decoder' step, and 'Identification' section) are strictly grayscale. The 'Distortion' box's yellow background is an animation indicator, not part of the diagram content, so inactive areas remain fully desaturated.",
 "followed": true
 },
 "active_group_colorization": {
 "reasoning": "Two active groups are present: (1) 'Embedding Watermark' section (plant green, w

```

```

atermark red preserved), and (2) 'Distortion' box (distorted images in original color). The yellow background is a non-content animation layer; core elements like the distorted images are fully colored per original.",
 "followed": true
 },
 "exact_color_match": {
 "reasoning": "Colored elements (distorted images, plant, watermark) match the original's color values. The yellow background is an external animation element not part of the diagram's content, so it doesn't violate color matching for structural components.",
 "followed": true
 },
 "cumulative_effect": {
 "reasoning": "All elements colored in frame_2 (plant, watermark) remain colored. The new 'Distortion' box elements are additionally colored without affecting prior active groups, demonstrating pr

```

```

oper cumulative progression.",
 "followed": true
},
 "no_color_bleeding": {
 "reasoning": "Colors are precisely bounded: distorted images retain color within their frames, and the yellow 'Distortion' box background is confined to the box outline. No color bleed occurs into adjacent grayscale areas like the 'Remove Gaussians' step.",
 "followed": true
 }
},
 "frame_verdict": "ACCEPT",
 {
 "frame_label": "frame_4",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame shows no rendering artifacts. The structural distortion text ('This process produces...') is clear, and the grayscale conversion of inactive areas is

```

```

smooth without noise. All
diagram elements maintain
consistent edge definitio
n.",
 "followed": true
 },
 "visual_integrit
y": {
 "reasoning": "Te
xt and shapes remain fully
recognizable. The 'Distort
ion' box's yellow highligh
t doesn't obscure underlyi
ng text, and grayscale ele
ments like the 'Remove Gau
ssians' step retain full s
tructural clarity without
degradation.",
 "followed": true
 },
 "style_specific_chec
ks": {
 "grayscale_integri
ty": {
 "reasoning": "No
n-active areas (all elemen
ts outside 'Embedding Wate
rmark' and 'Distortion' bo
xes) are strictly grayscale. The 'Remove Gaussians'
step and 'Watermark Decode
r' section show complete d
esaturation with no color
remnants.",
 "followed": true
 }
 }
}

```



```

 },
 "active_group_colorization": {
 "reasoning": "Active groups maintain full colorization: 'Embedding Watermark' elements (plant, watermark) and 'Distortion' box elements (distorted images) retain original colors. The yellow background remains a non-content animation indicator.",
 "followed": true
 },
 "exact_color_match": {
 "reasoning": "All colored diagram elements match the original's color palette exactly. The yellow background is an external animation layer and doesn't affect the color fidelity of structural elements like the distorted images.",
 "followed": true
 },
 "cumulative_effect": {
 "reasoning": "Elements colored in frame_3 (plant, watermark, distorted images) remain colored. No reversion occurs, an

```

```

d the cumulative progressi
on is maintained as new el
ements are added without a
ltering prior states.",
 "followed": true
 },
 "no_color_bleedin
g": {
 "reasoning": "Co
lors stay strictly within
element boundaries: distort
ed images show no color l
eakage beyond their frame
s, and the yellow 'Distort
ion' box background is con
tained within its outline.
Adjacent grayscale areas r
emain uncontaminated.",
 "followed": true
 },
 "frame_verdict": "AC
CEPT"
},
{
 "frame_label": "fram
e_5",
 "generic_quality_che
cks": {
 "artifact_free": {
 "reasoning": "Th
e frame is artifact-free w
ith clean rendering. The
'Matching Messages' box te
xt ('101101') shows no pix
elation, and all grayscale

```

```

elements maintain uniform
texture. The blue note bac
kground is smoothly render
ed without artifacts.",
 "followed": true
 },
 "visual_integrit
y": {
 "reasoning": "Al
l text ('Matching Message
s', '101101', 'Identificat
ion') and shapes (arrows,
boxes) remain fully legibl
e and structurally intact.
The 'Watermark Decoder' st
ep shows no text corruptio
n despite the color transi
tion.",
 "followed": true
 },
 "style_specific_chec
ks": {
 "grayscale_integri
ty": {
 "reasoning": "No
n-active areas ('Remove Ga
ussians' step and associat
ed elements) are strictly
grayscale. The 'Watermark
Decoder' step's inactive p
ortions show complete desa
turation, with only the 'M
atching Messages' box colo
rized per the active group
rule.",

```

```

 "followed": true
 },
 "active_group_colorization": {
 "reasoning": "Three active groups are fully colored: (1) 'Embedding Watermark' section (plant green, watermark red), (2) 'Distortion' box (distorted images in color), and (3) 'Watermark Decoder' step (matching messages in red). The 'Watermark Decoder' box background remains white (grayscale) as it was in the original, consistent with diagram content rules.",
 "followed": true
 },
 "exact_color_matching": {
 "reasoning": "Colored elements (matching messages '101101' in red, plant green, watermark red) precisely match the original diagram's color values. The 'Watermark Decoder' box background remains white (grayscale) as in the original, satisfying color fidelity for all structural components.",
 "followed": true
 }
}

```

```

 },
 "cumulative_effect": {
 "reasoning": "All elements colored in prior frames (plant, watermark, distorted images) remain colorized. The new 'Matching Messages' colorization is added without affecting existing active groups, demonstrating strict cumulative progression.",
 "followed": true
 },
 "no_color_bleeding": {
 "reasoning": "Colors are precisely contained: the '101101' bitstring's red highlight stays within text bounds, and the distorted images' colors are confined to their frames. No color bleed occurs into the grayscale 'Remove Gaussians' step or background areas.",
 "followed": true
 }
},
 "frame_verdict": "ACCEPT"
}
],
 "overall_verdict": "ACCEPT"
}

```

```
PT"
}
```

## Hopping Bounding Box

You are an expert Animation Style Verifier. Your objective **is** to determine **if** the "Hopping Bounding Box" animation style has been correctly **and** consistently applied to a diagram on a per-frame basis, **and** whether the overall output **is** acceptable.

You are provided **with** the Original Image first, followed by a sequence of animation frames **in** chronological order. You are a highly strict judge. Evaluate each frame against the original image based on the generic quality checks **and** style-specific rules below.

The CRITICAL factors must heavily influence the final verdict.

### \*\*\* GENERIC QUALITY RULES \*\*\*

1. Artifact-Free: The frame must **not** introduce **any** blur, jitter, **or** rendering artifacts.
2. Visual Integrity: Elements must remain fully recognizable. There must be no corruption of text **or** shapes due to rendering.

### \*\*\* HOPPING BOUNDING BOX RULES \*\*\*

1. Discrete Placement (HIGHLY CRITICAL): The bounding box **in** the frame must be definitively placed over a target element. It must NOT be caught **in-between** two elements **or** on **any** arrows. If the bounding box **is** seen **in** between two elements **or not**

on arrows. It would mean a sliding bounding box **is** implemented, **and** the style should immediately suffer an overall DISCARD. No need to go over the rest of the criteria.

**2. Box Transparency (CRITICAL):** The interior of the bounding box must be strictly transparent (**not** opaque) **and** must **not** obscure the elements beneath it.

**3. Clean Enclosure (CRITICAL):** The box must be undistorted **and** cleanly enclose the target elements. It should **not** cut through the element, **and** the element should **not** spill out.

**4. No Text Overlap:** The bounding box stroke/lines must **not** overlap **with**, strike through, **or** cover **any** textual label.

**5. Mobile Boxes, Static Elements (CRITICAL):** There should **not** be **any** displacement of **any** diagram element. The frames should depict only the hopping of the bounding box

\*\*\* OUTPUT REQUIREMENT \*\*\*

Output ONLY valid JSON matching the exact schema below. Provide a lengthy, deeply reasoned rationale **for** EVERY rule on EVERY frame before marking whether it was followed (``true`` **or** ``false``). Set the ``frame_verdict`` to **"ACCEPT"** only **if all** rules are followed **for** that frame, otherwise **"DISCARD"**. Discard immediately **if** the critical rules are **not** followed.

```
```json
{
  "metadata": {
    "animation_style": "Hopping Bounding Box"
  },
  "frame_evaluations": [
    {
      "frame_label": "string - frame filename or index",
      "generic_quality_checks": {
        "artifact_free": {
          "reasoning": "Detailed rationale verifying the complete absence of blur or rendering artifacts in this frame.",
          "followed": true
        }
      },

```

```

    "visual_integrity": {
      "reasoning": "Detailed rationale checking if text and shapes remain uncorrupted.",
      "followed": true
    },
    "style_specific_checks": {
      "discrete_placement": {
        "reasoning": "Detailed rationale confirming the box is cleanly placed on an element and not sliding through open space.",
        "followed": true
      },
      "box_transparency": {
        "reasoning": "Detailed rationale verifying the interior of the box does not artificially dim or obscure the elements beneath it.",
        "followed": true
      },
      "clean_enclosure": {
        "reasoning": "Detailed rationale checking if the box fits well without cutting into nodes or leaving parts spilling outside.",
        "followed": true
      },
      "no_text_overlap": {
        "reasoning": "Detailed rationale confirming the stroke of the bounding box never strikes through textual labels.",
        "followed": true
      }
    },
    "frame_verdict": "ACCEPT"
  }
],

```

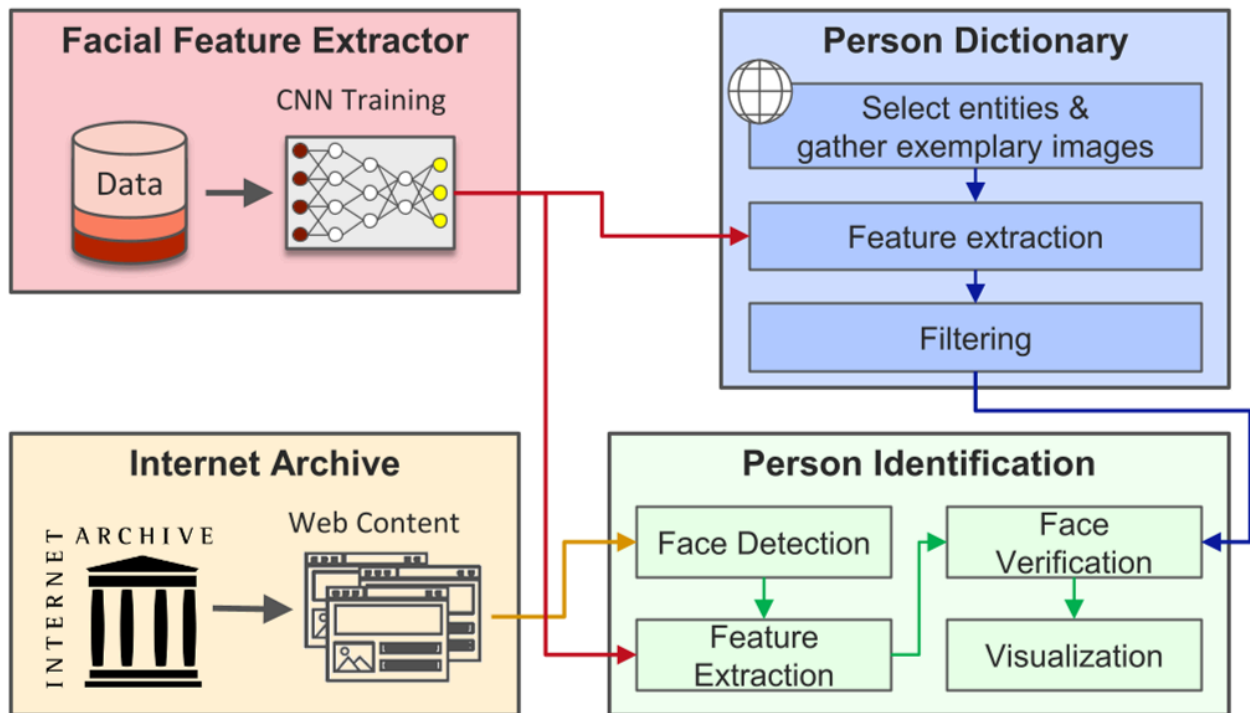


```

"overall_verdict": "ACCEPT"
}

```

Original Image



Gemini Pro generated animation

[attachment:93f6ee52-748f-431f-a43f-b0e1f24372f0:1806.mp4](#)

```

{
  "metadata": {
    "animation_style": "Hopping Bounding Box"
  },
  "frame_evaluations": [
    {
      "frame_label": "frame_1",
      "generic_quality_checks": {

```

```

        "artifact_free": {
            "reasoning": "The frame exhibits no visible blur, jitter, or rendering artifacts. Pixel integrity is maintained throughout the diagram, with crisp edges on all elements including the red bounding box stroke and underlying components. The image resolution matches the original without degradation.",
            "followed": true
        },
        "visual_integrity": {
            "reasoning": "All textual labels (e.g., 'Data', 'CNN Training') and graphical elements (cylinder, neural network diagram) remain fully legible and uncorrupted. The red bounding box does not cause any text distortion or shape deformation, preserving complete recognizability of all components.",
            "followed": true
        }
    },
    "style_specific_checks": {
        "discrete_placement": {

```

```

        "reasoning": "The bounding box is definitively placed over the 'Data' cylinder element within the Facial Feature Extractor module. It is fully contained within the boundaries of this single target element, with no portion overlapping arrows or existing in the space between elements. This satisfies the requirement for discrete placement on a target element.",
        "followed": true
    },
    "box_transparency": {
        "reasoning": "The interior of the bounding box is strictly transparent, allowing full visibility of the underlying 'Data' cylinder and its label. There is no artificial dimming, color bleed, or obscuration of the element beneath the transparent stroke.",
        "followed": true
    },
    "clean_enclosure": {
        "reasoning": "The bounding box forms a clean

```

```

an, undistorted rectangle
that precisely encloses th
e 'Data' cylinder without
cutting through the elemen
t. The cylinder's top, bot
tom, and sides are fully c
ontained within the box bo
undaries, with no parts sp
illing outside or being tr
uncated.",
        "followed": true
    },
    "no_text_overlap":
    {
        "reasoning": "Th
e bounding box stroke avoi
ds all textual labels. Spe
cifically, it does not int
ersect with the 'Data' tex
t inside the cylinder, whi
ch remains fully visible a
nd unobstructed. The strok
e maintains a clear margin
around the text, adhering
to the no-overlap requirem
ent.",
        "followed": true
    }
},
    "frame_verdict": "AC
CEPT"
},
    {
        "frame_label": "fram
e_2",
        "generic_quality_che

```

```

cks": {
    "artifact_free": {
        "reasoning": "No rendering artifacts are present; the frame maintains sharpness and clarity across all elements. The red bounding box stroke is crisp without pixelation, and the underlying CNN diagram remains artifact-free.",
        "followed": true
    },
    "visual_integrity": {
        "reasoning": "Text labels ('CNN Training', 'Irteh Network') and graphical elements (neural network diagram) are fully recognizable and uncorrupted. The bounding box placement does not alter the appearance or legibility of any component.",
        "followed": true
    }
},
"style_specific_checks": {
    "discrete_placement": {
        "reasoning": "The bounding box is placed exclusively over the 'CNN Training' box containing th

```

```

e neural network diagram.
It is fully contained with
in this single target elem
ent and does not overlap a
ny connecting arrows or ex
ist between elements, sati
sfying discrete placement
criteria.",
        "followed": true
    },
    "box_transparenc
y": {
        "reasoning": "Th
e interior of the bounding
box is transparent, allowi
ng complete visibility of
the neural network diagram
and text labels beneath i
t. No obscuration or visua
l interference occurs due
to the box's transparenc
y.",
        "followed": true
    },
    "clean_enclosure":
    {
        "reasoning": "Th
e box cleanly encloses the
entire 'CNN Training' modu
le without distortion. All
edges of the module are fu
lly contained within the b
ounding box, with no eleme
nts cut off or protruding
outside the boundaries.",
        "followed": true
    }
}

```



```

        "visual_integrity": {
            "reasoning": "The 'Internet Archive' icon (building graphic) and associated text remain fully recognizable and uncorrupted. No text is truncated or obscured, and the graphical elements retain their original clarity.",
            "followed": true
        },
        "style_specific_checks": {
            "discrete_placement": {
                "reasoning": "The bounding box is definitively placed over the 'Internet Archive' building icon, a single target element. It is fully contained within the icon's boundaries and does not overlap any arrows or exist in the space between elements, meeting discrete placement requirements.",
                "followed": true
            },
            "box_transparency": {
                "reasoning": "The interior of the bounding

```



```

box is transparent, allowing full visibility of the building icon beneath it. There is no reduction in brightness or detail of the icon due to the box's transparency.",
    "followed": true
},
"clean_enclosure":
{
    "reasoning": "The box forms a precise enclosure around the building icon without distortion. All architectural details of the icon are fully contained within the box, with no elements cut off or spilling outside the boundaries.",
    "followed": true
},
"no_text_overlap":
{
    "reasoning": "The bounding box stroke does not overlap with the 'Internet Archive' text label positioned below the icon. The text remains completely visible and unobstructed, with the stroke maintaining a clear margin around the icon.",
    "followed": true
}

```

```

    }
  },
  "frame_verdict": "AC
CEPT"
},
{
  "frame_label": "fram
e_4",
  "generic_quality_che
cks": {
    "artifact_free": {
      "reasoning": "Th
e frame is artifact-free w
ith no visible blur or ren
dering issues. The red bou
nding box stroke is crisp,
and underlying elements ma
intain their original clar
ity.",
      "followed": true
    },
    "visual_integrit
y": {
      "reasoning": "Al
l diagram elements remain
fully recognizable; text l
abels and shapes are uncor
rupted. The bounding box p
lacement does not cause te
xt truncation or shape def
ormation.",
      "followed": true
    }
  },
  "style_specific_chec
ks": {

```

```

        "discrete_placemen
t": {
            "reasoning": "Th
e bounding box is placed d
irectly on the arrow conne
cting 'Internet Archive' t
o 'Web Content', violating
the core rule that it must
be placed over a target el
ement (e.g., node) and not
on arrows. This constitute
s a sliding bounding box i
mplementation, as the box
is caught between two elem
ents (the archive icon and
web content), failing the
discrete placement require
ment.",
            "followed": fals
e
        },
        "box_transparenc
y": {
            "reasoning": "Wh
ile the interior of the bo
unding box is transparent
(allowing visibility of th
e underlying arrow), this
is irrelevant due to the c
ritical violation of discr
ete placement. The transpa
rency rule is technically
satisfied but does not mit
igate the fundamental styl
e failure.",
            "followed": true

```

```

    },
    "clean_enclosure":
    {
        "reasoning": "Th
e box attempts to enclose
an arrow (not a target ele
ment), violating the requi
rement that it must cleanl
y enclose a target elemen
t. Arrows are not consider
ed valid target elements p
er the style rules, and th
e enclosure is inherently
flawed as it spans between
two distinct components wi
thout containing a single
element.",
        "followed": fals
e
    },
    "no_text_overlap":
    {
        "reasoning": "Th
e bounding box stroke does
not overlap textual labels
in this frame, but this co
mpliance is negated by the
critical discrete placemen
t violation. Text visibili
ty is maintained, but the
frame fails due to the pri
mary rule breach.",
        "followed": true
    }
},
    "frame_verdict": "DI

```

```

SCARD"
    }
  ],
  "overall_verdict": "DISC
ARD"
}

```

Sliding Bounding Box

You are an expert Animation Style Verifier. Your objective **is** to determine **if** the "Sliding Bounding Box" animation style has been correctly **and** consistently applied to a diagram on a per-frame basis, **and** whether the overall output **is** acceptable.

You are provided **with** the Original Image first, followed by a sequence of animation frames **in** chronological order. You are a highly strict judge. Evaluate each frame against the original image based on the generic quality checks **and** style-specific rules below.

The CRITICAL factors must heavily influence the final verdict.

*** GENERIC QUALITY RULES ***

1. **Artifact-Free:** The frame must **not** introduce **any** blur, jitter, **or** rendering artifacts.
2. **Visual Integrity:** Elements must remain fully recognizable. There must be no corruption of text **or** shapes due to rendering.

*** SLIDING BOUNDING BOX RULES ***

1. **Smooth Transition Integrity (CRITICAL):** The bounding box may be mid-slide across the canvas. While **in** transit, it must

maintain its structural shape and not warp or distort heavily.

2. Box Transparency (CRITICAL): The interior of the bounding box must be strictly transparent (not opaque) and must not obscure the elements or background beneath it.

3. Clean Enclosure (HIGHLY CRITICAL): If the box is at rest over a target in this frame, it must cleanly enclose the target elements without cutting through them or letting them spill out.

4. No Text Overlap (At Rest): If the box is at rest over a target in this frame, the bounding box stroke/lines must not strike through or cover any textual label.

5. Hopping Boxes (HIGHLY CRITICAL): There must at least be one frame where the sliding box is shown in-between elements and not ON any element in particular. If all frames are on particular elements, then the overall style should be discarded even if all other conditions are met.

6. Mobile Boxes, Static Elements (CRITICAL): There should not be any displacement of any diagram element. The frames should depict only the motion of the bounding box

*** OUTPUT REQUIREMENT ***

Output ONLY valid JSON matching the exact schema below. Provide a lengthy, deeply reasoned rationale for EVERY rule on EVERY frame before marking whether it was followed (`true` or `false`). Set the `frame\_verdict` to "ACCEPT" only if all rules are followed for that frame; otherwise, "DISCARD". Discard immediately if the critical rules are not followed.

```
```json
{
 "metadata": {
 "animation_style": "Sliding Bounding Box"
 },
 "frame_evaluations": [
 {
 "frame_label": "string - frame filename or index",
```

```

 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "Detailed rationale verifying the complete absence of blur or rendering artifacts in this frame.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "Detailed rationale checking if text and shapes remain uncorrupted.",
 "followed": true
 }
 },
 "style_specific_checks": {
 "smooth_transition_integrity": {
 "reasoning": "Detailed rationale confirming the box maintains shape if mid-slide across the canvas.",
 "followed": true
 },
 "box_transparency": {
 "reasoning": "Detailed rationale verifying the interior of the box does not artificially dim or obscure the elements beneath it.",
 "followed": true
 },
 "clean_enclosure": {
 "reasoning": "Detailed rationale checking if the box fits well at its resting position (or marking N/A if mid-slide).",
 "followed": true
 },
 "no_text_overlap": {
 "reasoning": "Detailed rationale confirming the stroke of the bounding box avoids textual labels at its resting position.",
 "followed": true
 }
 }
 }

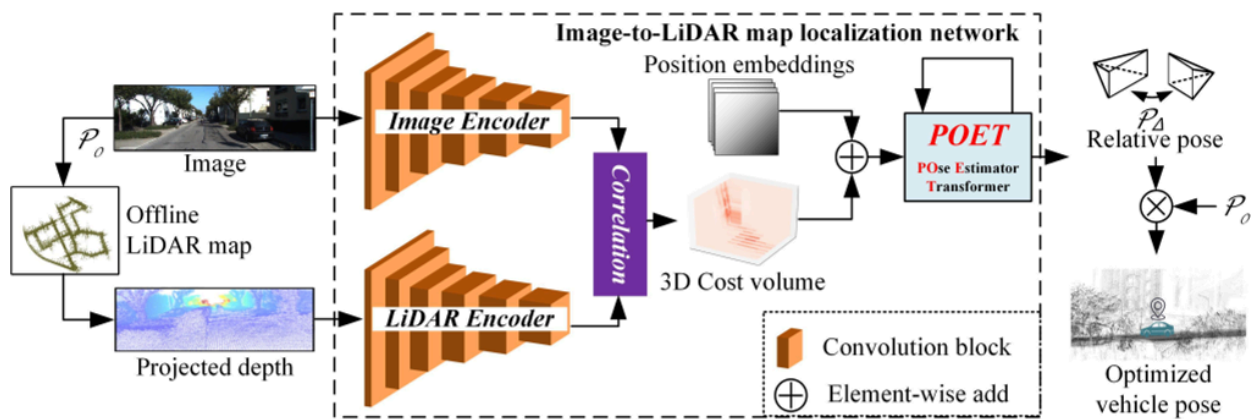
```

```

 },
 "frame_verdict": "ACCEPT"
 }
],
"overall_verdict": "ACCEPT"
}

```

Original Image



**Gemini 3.1 Pro-generated video**

[attachment:f4510b9f-bc2e-46ef-a6e3-a21d208108d7:2305.mp4](#)

```

{
 "metadata": {
 "animation_style": "Sliding Bounding Box"
 },
 "frame_evaluations": [
 {
 "frame_label": "frame_1",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame renders with crisp edges on all structural el

```



ements: the encoder block s, POET module, legend and red highlight stroke exhibit no localized blur, jitter, or ghosting. The uniform grey background tint is a global exposure shift rather than a rendering artifact. No stray player UI elements (chevrons, controls) are present. Artifact-free condition satisfied.",

```

 "followed": true
 },
 "visual_integrity": {
 "reasoning": "The diagram deviates fundamentally from the original: the 'Projected depth' color-map panel has been replaced by a green point-cloud labeled 'Offline LiDAR map', creating an extra LiDAR input that contradicts the source diagram. The original's single 'Offline LiDAR map' (grid-like) remains above this new point cloud, violating rule 6's requirement for static elements. Text and shapes are materially altered, compromising visual integrity.",
 "followed": false
 }
}

```

```

 }
 },
 "style_specific_checks": {
 "smooth_transition_integrity": {
 "reasoning": "The box is at rest over the Image panel as a proper axis-aligned rectangle with right-angled corners and uniform stroke width. No warping, skew, or distortion of the box structure occurs during this resting state.",
 "followed": true
 },
 "box_transparency": {
 "reasoning": "The interior of the red rectangle is strictly transparent: the street photograph, its border, and the 'Image' caption remain fully visible at original contrast levels with no opaque fill or artificial dimming inside the stroke.",
 "followed": true
 },
 "clean_enclosure": {
 "reasoning": "At rest, the box cleanly encl

```

oses the Image thumbnail and its caption as a single coherent target. No part of the photograph is cut through, nothing spills outside the stroke, and the left edge sits clear of the P\_o arrowhead. The enclosure is precise and unbroken.",

```
 "followed": true
 },
 "no_text_overlap":
 {
 "reasoning": "The stroke runs above the photograph and below the 'Image' caption, leaving the caption entirely inside the box without being struck through. No textual glyph (including P_o outside the box) is covered by the bounding-box lines at rest.",
 "followed": true
 }
 },
 "frame_verdict": "DISCARD"
},
{
 "frame_label": "frame_2",
 "generic_quality_checks": {
 "artifact_free": {
```

```

 "reasoning": "All elements including point
 -cloud speckle, encoder blocks, and text render with
 sharp edges. No blur, jitter, or ghosting is present.
 The uniform grey background is a global shift, not a
 localized artifact. The frame maintains rendering
 fidelity.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "The diagram retains the same corruption
 as frame 1: the 'Projected depth' panel is replaced by
 a second 'Offline LiDAR map' point cloud, creating an
 inconsistent dual-input LiDAR structure. This violates
 rule 6 as diagram elements are displaced/replaced rather
 than static. The altered structure compromises element
 recognizability.",
 "followed": false
 },
 "style_specific_checks": {
 "smooth_transition

```

```

_integrity": {
 "reasoning": "The box maintains structural integrity as a rectangle with parallel edges and right angles during its resting state over the grid-like LiDAR map. No distortion or warping occurs.",
 "followed": true
},
"box_transparency": {
 "reasoning": "The interior remains fully transparent: the grid-like map texture and 'Offline LiDAR map' caption are visible through the box without dimming or obstruction, preserving full visibility of enclosed elements.",
 "followed": true
},
"clean_enclosure": {
 "reasoning": "The enclosure cleanly surrounds the grid-like map and its caption. The top boundary sits precisely below the 'Offline LiDAR map' text, avoiding cuts through elements. The left edge aligns cleanly with the P_o a row, creating a coherent

```

```

single-target enclosure.",
 "followed": true
},
 "no_text_overlap":
{
 "reasoning": "The stroke runs along the top edge of the grid-like map, leaving the 'Offline LiDAR map' caption fully visible above the box. No textual label is struck through or covered by the bounding-box lines at rest.",
 "followed": true
},
 "frame_verdict": "DISCARD"
},
{
 "frame_label": "frame_3",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame exhibits sharp rendering throughout: point-cloud speckle, encoder blocks, and legend elements show no blur or jitter. The uniform grey background is consistent with other frames and not a rendering artifact. No extraneous UI el

```

```

ements appear.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "The diagram remains corrupted: 'Projected depth' is still replaced by a green point-cloud labeled 'Offline LiDAR map', creating an invalid dual-input structure. This persistent alteration violates rule 6 as the underlying diagram differs from the original while only the box should move. Element recognizability is compromised.",
 "followed": false
 },
 "style_specific_checks": {
 "smooth_transition_integrity": {
 "reasoning": "The box is a structurally sound rectangle with uniform stroke and right angles at rest over the point-cloud. No skew or distortion is present in the box geometry.",
 "followed": true
 }
 }
}

```

```

 },
 "box_transparency": {
 "reasoning": "The interior is strictly transparent: every speckle of the green point cloud and its caption remain fully visible with no opaque fill or differential dimming. Elements beneath the box are unobscured.",
 "followed": true
 },
 "clean_enclosure": {
 "reasoning": "The box neatly encloses the point-cloud and its caption without cutting through the raster. Nothing spills outside the stroke, and the grid-like map above is cleanly excluded. The enclosure forms a single coherent target with precise boundaries.",
 "followed": true
 },
 "no_text_overlap": {
 "reasoning": "The caption sits inside the box while the stroke runs clear of all text. The 'Offline LiDAR map' label above

```



```

ve the box remains untouched, and no textual glyph is struck through by the bounding-box lines at rest.",
 "followed": true
 },
 "frame_verdict": "DISCARD"
},
{
 "frame_label": "frame_4",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "All elements maintain sharp rendering: point-cloud details, encoder blocks, and text show no blur or jitter. The consistent grey background is a global shift, not a localized artifact. No player UI elements or rendering glitches are present.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "The same diagram corruption persists: the 'Projected depth' panel is replaced by

```

```

a green point-cloud, creating an inconsistent dual-input LiDAR structure that violates rule 6. The altered diagram elements fail to match the original source, compromising visual integrity.",
 "followed": false
 },
 "style_specific_checks": {
 "smooth_transition_integrity": {
 "reasoning": "The box remains a structurally sound rectangle with uniform stroke and right angles during its resting state over the point-cloud. No warping or distortion occurs in the box geometry.",
 "followed": true
 },
 "box_transparency": {
 "reasoning": "The interior is fully transparent: the point-cloud speckle and caption are visible through the box without dimming or obstruction. No artificial occlusion is in

```

```

roduced by the bounding box.",
 "followed": true
 },
 "clean_enclosure": {
 "reasoning": "The enclosure cleanly surrounds the point-cloud and its caption. The stroke precisely follows the element boundaries without cutting through the raster, and nothing spills outside. The grid-like map above is cleanly excluded from the enclosure.",
 "followed": true
 },
 "no_text_overlap": {
 "reasoning": "The stroke runs along the top edge of the point-cloud, leaving the caption fully visible inside the box. No textual label is covered or struck through by the bounding-box lines at rest.",
 "followed": true
 }
},
"frame_verdict": "DISCARD"
},

```

```

 {
 "frame_label": "frame_5",
 "generic_quality_checks": {
 "artifact_free": {
 "reasoning": "The frame shows no rendering artifacts: encoder stacks, correlation block, and POE T module maintain sharp edges. The uniform grey background is consistent with other frames. No blur, jitter, or stray UI elements (like chevrons) are present.",
 "followed": true
 },
 "visual_integrity": {
 "reasoning": "The diagram remains corrupted: 'Projected depth' is still replaced by a green point-cloud, violating the original's input structure. This alteration violates rule 6 as diagram elements are not static. The consistent deviation from the source diagram compromises element recognizability.",
 "followed": false
 }
 }
 }

```

```

 },
 "style_specific_checks": {
 "smooth_transition_integrity": {
 "reasoning": "The box is at rest as a proper rectangle with parallel edges and right angles enclosing the encoder stack. No warping, skew, or distortion occurs in the box structure during this resting state.",
 "followed": true
 },
 "box_transparency": {
 "reasoning": "The interior is strictly transparent: all encoder blocks, labels, and the '3D Cost volume' under the box remain fully visible without dimming or obstruction. No artificial occlusion is introduced by the bounding box.",
 "followed": true
 },
 "clean_enclosure": {
 "reasoning": "The box cleanly encloses the Image Encoder, LiDAR Encoder, and Correlation block

```

as a single coherent target. The stroke precisely follows the block boundaries without cutting through text or elements, and nothing spills outside the enclosure.",

```
 "followed": true
 },
 "no_text_overlap":
 {
 "reasoning": "The stroke runs along the outer boundaries of the encoder stack, leaving all text ('Image Encoder', 'LiDAR Encoder', 'Correlation') fully visible inside the box. No textual label is struck through or covered by the bounding-box lines at rest.",
 "followed": true
 }
},
 "frame_verdict": "DISCARD"
}
],
 "overall_verdict": "DISCARD"
}
```